Lab 1: Requirements Engineering & UML Use-Case Modelling

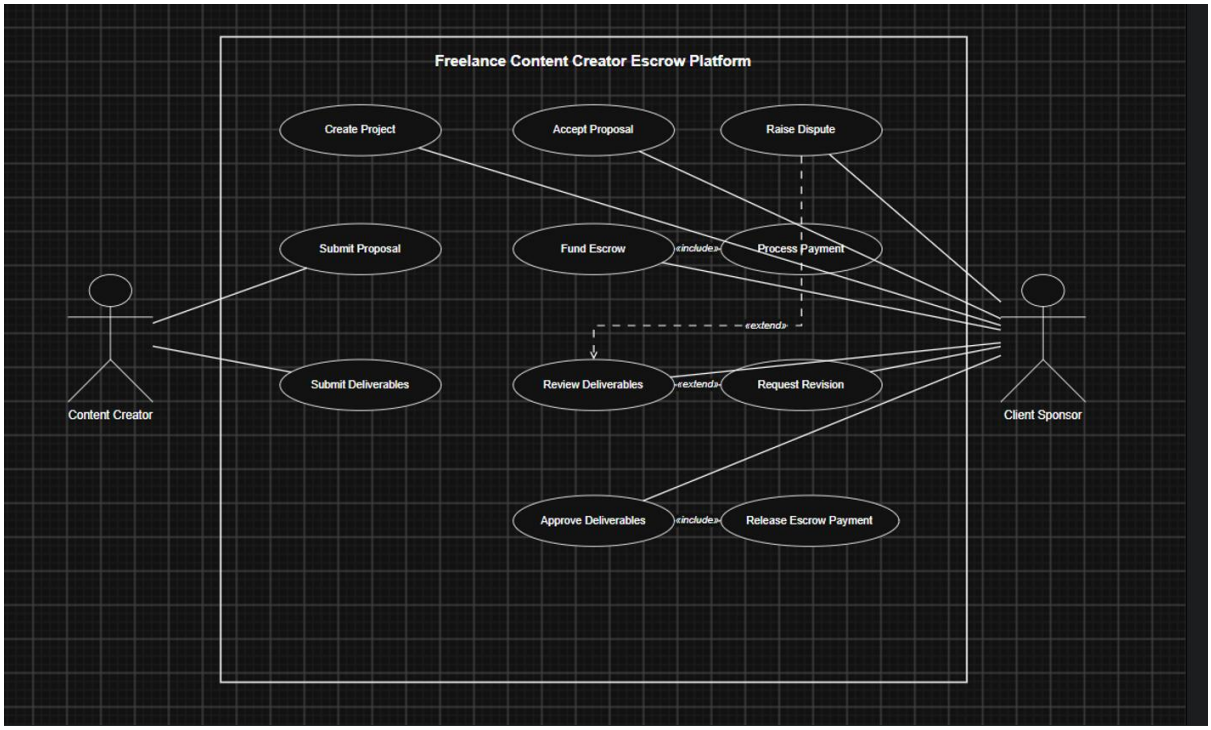
Problem No.: 53
Problem Statement: Freelance Content Creator Escrow Platform
Domain: Media, Events & Community
Name: L A GOKUL
SRN: PES2UG24CS246
Actors: Content Creator, Client Sponsor

1. Requirements Table :

	A	B	C	D	E	F	G	H
1	**1: Requirements Engineering & UML Use-Case Modelling**							
2	**em No.**					53		
3	**em Statem**	Freelance Content Creator Escrow Platform						
4	**ain**	Media, Events & Community						
5	**e**	L A GOKUL						
6		PES2UG24CS246						
7	**s**	Content Creator, Client Sponsor						
8								

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
1	Functional	The system shall allow the Client Sponsor to create a project by specifying content requirements, deliverables, deadline, and agreed payment amount.	High	Pass: Project is created when all mandatory details are provided. Fail: Project is created with missing mandatory details.	Provides the basic mechanism for initiating a freelance content project.
2	Functional	The system shall allow the Content Creator to submit a proposal for a project and allow the Client Sponsor to accept the proposal.	High	Pass: Proposal is submitted successfully and accepted proposals change the project status to Accepted. Fail: Proposal cannot be submitted or accepted correctly.	Establishes the agreement between the Content Creator and Client Sponsor.
3	Functional	The system shall allow the Client Sponsor to deposit the agreed project payment into an escrow account before the Content Creator begins the project.	High	Pass: Successful payment changes the escrow status to Funded. Fail: Escrow is marked funded without successful payment.	Secures the agreed payment before work begins.
4	Functional	The system shall allow the Content Creator to submit project deliverables and the Client Sponsor to review them and request revisions when required.	High	Pass: Submitted deliverables are stored and available for Client Sponsor review. Fail: Submitted deliverables cannot be accessed for review.	Provides a controlled content submission and review process.
5	Functional	The system shall release the escrowed payment to the Content Creator after the Client Sponsor approves the submitted deliverables.	High	Pass: Approval changes payment status to Released and project status to Completed. Fail: Payment is released without approval or remains unreleased after valid approval.	Completes the project transaction while protecting both parties.
01	Nonfunctional	The system shall display updated project and escrow transaction status within 5 seconds under normal operating conditions.	High	Pass: At least 95 of 100 test transactions display the updated status within 5 seconds. Fail: More than 5 transactions exceed 5 seconds.	Ensures responsive project and payment status updates.
02	Nonfunctional	The system shall protect project, user, and escrow transaction information using authenticated access and encrypted communication.	High	Pass: Unauthenticated users cannot access protected information and communication uses HTTPS/TLS. Fail: Protected information is accessible without authentication or encryption.	Protects sensitive project and financial information.

2. UML Use-Case Diagram:



3.Use-Case Flow Specification :

Use Case ID: UC-03
Use Case Name: Fund Escrow
Primary Actor: Client Sponsor

Preconditions :

1. The Client Sponsor has an active account and is authenticated.
2. A project has been created.
3. The Content Creator's proposal has been accepted.
4. The agreed project payment amount is available.

Postconditions :

1. The escrow payment is successfully processed.
2. The escrow status is changed to **Funded**.
3. The Content Creator can begin work on the project.
4. The transaction is recorded securely.

Main Success Scenario :

1. Client Sponsor opens the accepted project.
2. System displays the agreed payment amount.
3. Client Sponsor selects **Fund Escrow**.
4. System displays the payment details and requests payment confirmation.
5. Client Sponsor confirms the payment.
6. System sends the payment request for processing.
7. Payment is successfully authorized.
8. System deposits the agreed amount into the escrow account.
9. System updates the escrow status to **Funded**.
10. System displays a successful escrow funding confirmation.
11. Use case ends successfully.

Alternate Flow

A1. Payment Failure :

1. At Step 7, if the payment is declined or fails, the system does not mark the escrow as funded.
2. System displays a payment failure message.
3. Client Sponsor is given an option to retry the payment.
4. If the retry is successful, the system proceeds with Step 8 of the Main Success Scenario.

5. If the payment continues to fail, the escrow remains **Unfunded** and the use case ends.